

SCI-BEAM

TolTEC Beammap Analysis

Effective PSF, Detector Calibration, Sensitivity, and APT Production

Scientific contract version v0.1

Science-team rationale revision r0.3

2026-08-17

A scientist-facing explanation of what a TolTEC Beammap measures, how BEAM derives the Beammap APT, what its effective PSF, `flxscale`, and `sens` mean, and what evidence is still required. The separate Formal Scientific/Engineering Contract is the normative conformance authority.

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Executive summary and current status

SCI-BEAM turns a Beammap observation into a scientifically traceable detector property table. Its primary measurement is not an ideal optical beam. It is the observation-local effective point-spread-function (PSF) core visible in each standardized per-detector Beammap after that map's sampling, pixelization, projection, gridding, filtering, finite boundary, and other realized response. BEAM also measures the source amplitude in the raw detector observable, derives relative focal-plane coordinates, calibrates each detector to a fixed nominal-beam top-of-atmosphere convention, estimates an NEFD-like sensitivity, and publishes the mandatory Beammap APT plus the empirical maps and diagnostics needed to judge those summaries.

For one detector the physical input signal is

$$x_d(t) = \frac{\Delta f_d(t)}{f_d}, \quad (1)$$

the dimensionless fractional KID resonance-frequency shift. The fitted map $x_{d,p}^{\text{BM}}$ remains in $\Delta f/f$; **xs** denotes the collection of these raw detector signals, never a calibrated mJy beam⁻¹ product.

Input	Standardized detector-resolved Beammap in $\Delta f/f$, physical WCS/metric, valid support, covariance state, parent map-response lineage, calibrator/source model, fixed nominal beam, source atmosphere, and optional soft locator prior.
Output	Complete Beammap APT with raw/horizon coordinates, full effective-core tensor, amplitude, top-of-atmosphere nominal-beam flxscale , NEFD-like sens , independent quantity states, and immutable links to maps, residuals, covariance, and diagnostics.
Organizing fit	$x_{d,p}^{\text{BM}} = A_d^{\text{obs}} T(\boldsymbol{\vartheta}_p - \boldsymbol{\mu}_d; \boldsymbol{\Sigma}_{\text{eff},d}, \boldsymbol{\phi}) + \mathbf{b}_p^{\text{T}} \boldsymbol{\beta}_d + r_{d,p}$.
Calibration	The source flux $F_{s,a}^{\text{TOA,nom}}$ is required in top-of-atmosphere mJy per fixed nominal beam; $\text{flxscale}_d = F_{s,a}^{\text{TOA,nom}} / (C_{s,a}^{\text{eff}} A_d^{\text{obs}})$.
Status	Scientific authority v0.1/r0.3 frozen. Implementation, map-response fidelity, observational performance, science-impact qualification, and production readiness are not assessed.

The contract separates validity by quantity. A valid effective-core fit may coexist with unavailable **flxscale**; a valid calibration may coexist with unavailable **sens**; a precise Gaussian core may coexist with unknown wing power; and a useful telescope-health measurement need not be qualified as a downstream kernel.

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1 What BEAM measures

A Beammap deliberately moves a bright source through a detector field so that each detector obtains a spatial response measurement. SCI-BEAM v0.1 fits the standardized detector map, not the detector timestream. The map is required to remain in raw $\Delta f/f$ because the same observation supplies the evidence from which detector calibration is derived. A map already calibrated to mJy beam⁻¹ would obscure or circularize that derivation.

The empirical per-detector map is the measurement. The fitted ellipse is a compact model of its measurable high-signal-to-noise core. This distinction is essential for an aberrated telescope. Broad or asymmetric power may lie below the detector-map noise, beyond the mapped radius, or partly within a fitted background. A numerically precise core does not prove that the complete PSF is Gaussian, that its wings are absent, or that an intrinsic optical beam has been recovered.

Four uses therefore remain separate:

- **Amplitude calibration** asks whether the reference-origin source response is robust enough to support `flxscale`.
- **Sensitivity** asks what off-source raw-signal noise corresponds to in a top-of-atmosphere nominal-beam point-source convention.
- **Telescope-health diagnosis** asks how the effective core and residual response differ from the nominal expectation.
- **Kernel qualification** asks whether a Gaussian or empirical response is accurate enough for a named downstream scientific calculation.

Success for one use is not success for the others. A Beammap can be a valid health diagnostic precisely because it shows a severely aberrated response. Conversely, an apparently nominal core can still leave the complete response poorly constrained.

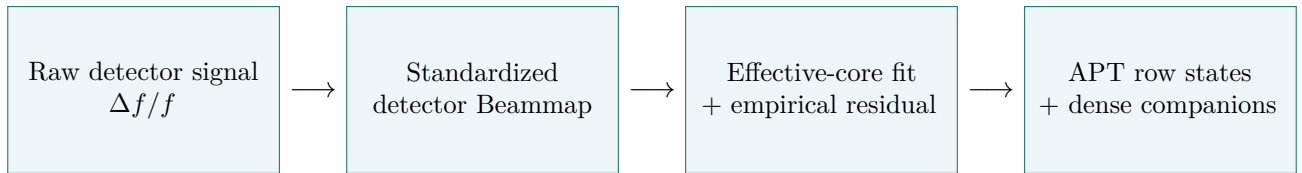


Figure 1: The v0.1 measurement chain. The effective PSF core is conditioned on the response embodied in the parent Beammap; the empirical map and residual remain part of the scientific result.

2 Nominal beam and calibrator model

Calibration needs a stable reference that is not redefined by the very Beammap being measured. For array a , the source model therefore supplies $F_{s,a}^{\text{TOA,nom}}$: the required calibrator source flux at the top of the atmosphere, in mJy per fixed versioned nominal beam, for the declared source, epoch, passband, spectrum, finite-source convention, and source-reference origin. The nominal beam is a calibration reference, not a claim about the instantaneous telescope PSF.

The source-brightness shape S is kept separate from the effective Gaussian core G . It is non-negative, unit-integral, and centered by a declared zero first moment. Their convolution is

$$H = S * G, \quad T(\mathbf{u}) = \frac{H(\mathbf{u})}{H(\mathbf{0})}, \quad T(\mathbf{0}) = 1. \quad (2)$$

The fitted amplitude is consequently the modeled source excess at the translated reference origin. For an asymmetric source that origin need not be the profile maximum. This convention preserves the subtle and

useful r0.1 finite-source treatment while making the calibration numerator explicit.

$F_{s,a}^{\text{TOA,nom}}$ is supplied in $\text{mJy beam}_{\text{nom}}^{-1}$ under the same source-reference-origin convention as the fitted amplitude. That declared quantity enters the calibration ratio directly.

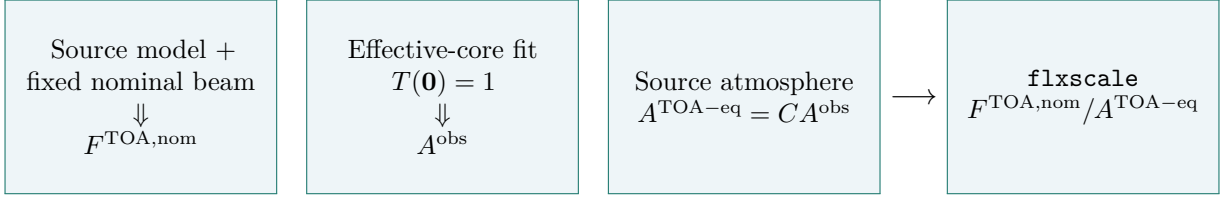


Figure 2: The amplitude convention. The fixed nominal-beam source flux and the fitted map amplitude use the same source-reference origin.

3 Per-detector Beammap fit

The fit uses a metrically orthonormal angular tangent plane. A full WCS transformation, or a validated local Jacobian, must establish the two angular scales, signs, handedness, rotation, shear or cross terms, and the correct azimuth metric. Raw ΔAz and ΔEl are not automatically a Euclidean plane. If a local linearization is used, its reference location and error across the entire fit region are part of the result.

The effective core is one symmetric positive-definite tensor,

$$\Sigma_{\text{eff}} = \mathbf{R}(\theta) \begin{pmatrix} \sigma_{\text{maj}}^2 & 0 \\ 0 & \sigma_{\text{min}}^2 \end{pmatrix} \mathbf{R}(\theta)^{\text{T}}. \quad (3)$$

All three independent shape degrees of freedom are fitted together. Separate axis-aligned one-dimensional widths followed by a rotation estimate would not be equivalent. At circularity the tensor and common width remain meaningful, but an orientation does not.

Scientists commonly use

$$\text{FWHM}_{\text{maj,min}} = 2\sqrt{2\ln 2} \sigma_{\text{maj,min}}. \quad (4)$$

When the Gaussian interpretation is available, the fitted core has solid angle $\Omega_G = 2\pi\sigma_{\text{maj}}\sigma_{\text{min}}$. This is not promoted to complete beam solid angle while material wings may be unmeasured.

The fit is joint. Its local model Jacobian contains derivatives with respect to amplitude, centroid, all tensor degrees of freedom, background, admitted source-shape nuisance parameters, and coordinate/WCS nuisance parameters when they are included jointly. The fit covariance retains material cross terms among those map-model parameters, including amplitude–width, centroid–shape, width–orientation, background–centroid, source-size–beam-width, and coordinate–shape relationships. Independent one-parameter error bars do not describe this problem. At circularity, uncertainty is expressed in an invariant tensor basis rather than as an arbitrary angle error.

Separate derived-product Jacobians propagate the calibrator source flux, nominal-beam convention, source atmosphere, source-model systematic terms, scan-noise statistic, and exposure normalization into **flxscale** and **sens**. The complete BEAM result retains material dependence between the map fit and these derived stages without relabeling every term as local fit covariance.

Support is part of the estimator. Cropping, a masked core, a disconnected region, or a one-sided background can change which quantities are identifiable. A successful optimizer cannot manufacture information absent from the map. The complete empirical map, support, fitted model, residual, background, and model-adequacy diagnostics are therefore preserved beside the compact APT summary.

A compatible TolAPT prior may seed or bound the source search, but it does not enter the measurement objective as hidden truth and cannot veto a blind search. The selected candidate, prior route, and fallback remain explicit. Optimizer completion alone is not scientific convergence: candidate identity, available parameters, support, objective, and valid-detector membership must be stable under the declared policy.

4 Detector coordinates and LMT field rotation

SCI-BEAM produces two relative coordinate representations. If $\mu_{d,\text{fit}}$ is the fitted source centroid, the raw detector coordinate is

$$\mathbf{x}_{t,d}^{\text{raw}} = \mathbf{S}_{\mu \rightarrow x,o} \mu_{d,\text{fit}}, \quad (5)$$

where $\mathbf{S}_{\mu \rightarrow x,o}$ is the declared sign and frame transformation for the parent Beammap. It may not be inferred later from implementation behavior. The horizon coordinate (x_t, y_t) is the same detector geometry after derotation into the declared horizon-fixed focal-plane frame.

The effective angle for detector d is derived from the parent-sample time and elevation support and contribution weights that produced the fitted map pixels, restricted through the realized pixel-domain fit support. Pixel fit weights alone are not parent timing evidence. The PSF tensor is rotated with the coordinate.

A single observation-average angle is acceptable only when its induced coordinate and tensor error is negligible relative to the required accuracy. Otherwise the transformation must preserve time dependence or a detector-specific effective angle. Each detector records the effective angle, time and elevation support, weighting rule, represented angle range, rotation-law version, and uncertainty.

The APT origin and the physical field-rotation pivot are different concepts. Relative detector coordinates are defined only up to a common translation; there is no scientifically robust reference detector. BEAM records a deterministic recentering rule, but that origin is a gauge choice, not a boresight measurement.

Likewise, the physical point about which the LMT field rotates is not yet established as telescope truth. The relation is affine rather than guaranteed to be a pure rotation about a known boresight. A conventional pivot may be used, but it is labeled conventional. Unknown pivot error appears principally as an elevation-dependent common pointing term and, when detectors have different effective angles, as smaller differential geometry uncertainty. BEAM does not relabel its ensemble translation as telescope pointing error, absolute array position, or a pointing-model correction.

This limitation makes one transfer rule essential: the bracketing pointing observations and the associated science observation must use the same immutable APT artifact and AST rotation convention. Then arbitrary centering and conventional-pivot terms are treated consistently and largely cancel. Changing the APT artifact between pointing and science is invalid unless a separately authorized coordinate transform proves equivalence.

A pointing observation and correction immediately before the Beammap normally place the source close to the expected location, but do not establish the physical field-rotation pivot.

5 Amplitude and top-of-atmosphere flxscale

A_d^{obs} is measured on the source-observation plane. BEAM owns the source-atmosphere correction used to place that amplitude on its top-of-atmosphere-equivalent plane:

$$A_d^{\text{TOA-eq}} = C_{s,a}^{\text{eff}} A_d^{\text{obs}}, \quad \text{flxscale}_d = \frac{F_{s,a}^{\text{TOA,nom}}}{A_d^{\text{TOA-eq}}} = \frac{T_{s,a}^{\text{eff}} F_{s,a}^{\text{TOA,nom}}}{A_d^{\text{obs}}}. \quad (6)$$

The factor has unit $\text{mJy beam}_{\text{nom}}^{-1}/(\Delta f/f)$ and retains the physical identity of the dimensionless detector observable. If the standardized map is already on a properly documented TOA-equivalent raw-signal plane, that route may be used instead; the route is always explicit.

The source atmosphere is applied exactly once in BEAM. SCI-CAL later applies the target-observation atmosphere exactly once to target samples. The source atmosphere is neither omitted from Beammap calibration nor repeated during target calibration. Source-model, nominal-beam, atmosphere, and fitted-amplitude uncertainties are often shared across detectors or arrays and must retain their correlation scope. BEAM owns the internal sequence from fitted amplitude to calibration candidate to scientifically accepted source-APT **flxscale**. It does not await a SCI-CAL promotion. TolProj may later associate detectors or apply an approved child-APT transformation. SCI-CAL consumes the selected observation-specific factor and applies it once.

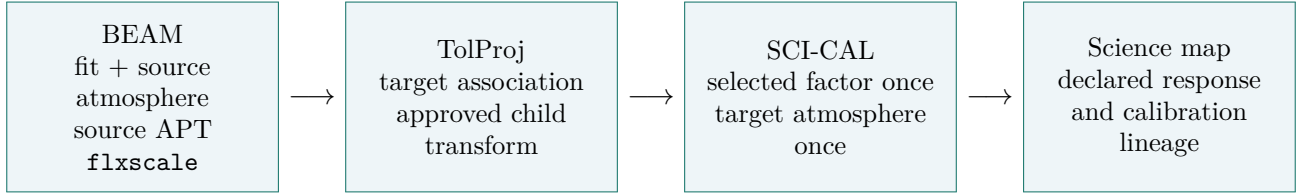


Figure 3: Producer–transformer–consumer lineage. BEAM owns source calibration; TolProj owns an approved relationship or transformation; SCI-CAL applies the selected factor and the target atmosphere without reinterpreting either.

6 NEFD-like sensitivity

The PSF fit is map-domain, but **sens** uses scan-domain data because it is a time-normalized noise measurement. For each detector, the actual detector trajectory identifies scans that do not pass within the declared source- exclusion region. A declared scan-level statistic $n_{d,s}$ is computed from each admitted off-source scan, placed on the same source-atmosphere convention as the APT calibration, and robustly summarized:

$$n_d^{\text{off}} = \text{median}_{s \in \mathcal{S}_{d,\text{off}}} n_{d,s}^{\text{TOA-eq}}, \quad \mathbf{sens}_d = |\mathbf{flxscale}_d| n_d^{\text{off}}. \quad (7)$$

With the declared time normalization, **sens** has units $\text{mJy beam}_{\text{nom}}^{-1} \sqrt{\text{s}}$. It is a per-detector, top-of-atmosphere, nominal-beam, NEFD-like point-source sensitivity and is finite and strictly positive whenever available. The sign of **flxscale** remains part of the detector response convention; only its magnitude enters sensitivity.

The quantity has two legitimate uses: a rough relative-sensitivity input for approximate weighting and a detector/array/system-health diagnostic. It is not a signal calibration factor, total calibrated uncertainty, exact inverse variance, empirical significance, or complete noise model. A weighting owner may adopt a relation proportional to $1/\mathbf{sens}^2$, but owns normalization, admission, covariance assumptions, and permitted claims.

The exact scan statistic is deliberately not invented in r0.3. Source- exclusion radius, baseline treatment, time-frequency range or bandwidth, robust or spectral estimator, correlated-noise treatment, minimum scan count, exposure normalization, partial-scan policy, and uncertainty/repeatability statistic require owner decisions and validation. Individual scan estimates and their scatter remain available. Broad diffuse aberrated response may also contaminate apparently off-source scans, so **sens** remains an effective point-source/system measure rather than pure intrinsic detector noise.

7 Aberration, adequacy, and broadening

Relative to a compatible fixed nominal Gaussian tensor, BEAM may form

$$\Sigma_{\text{broad}} = \Sigma_{\text{eff}} - \Sigma_{\text{nom}}. \quad (8)$$

This is a telescope-health diagnostic. Its eigenvectors describe broadening, not necessarily the major-axis orientation of the total effective PSF. A literal Gaussian-smoothing interpretation requires the difference to be positive semidefinite within uncertainty. An indefinite tensor is not clipped into a plausible kernel; it can indicate noise, convention mismatch, or failure of a nominal-plus-Gaussian-broadening model.

Model adequacy cannot be compressed into optimizer status. Quantitative diagnostics must examine structured residuals, support dependence, peak displacement, asymmetry, excess response outside the core, multiple peaks or lobes, core-versus-wing completeness, and possible amplitude bias. Thresholds will come from science-impact evidence, not from the Gaussian equations.

Array-, network-, or observation-level stacks are useful optional diagnostics for common diffuse response below individual-detector sensitivity. They must declare recentering, inclusion, weighting, normalization, coverage, detector variation, covariance, and parent maps. They are neither required APT fields nor prerequisites for per-detector core, `flxscale`, or `sens`, and they do not become kernels without separate qualification.

8 Beammap APT and companion products

The Beammap APT is a required SCI-BEAM output, not an optional convenience. Its per-detector scientific content resolves stable identity; raw and horizon coordinates plus coordinate covariance; effective angle; complete effective- core tensor, FWHM, and orientation state; raw amplitude and uncertainty; `flxscale`; `sens`; and independent availability, validity, uncertainty, and cause state for every interpretation and use.

The tensor meaning is authoritative even if storage later chooses equivalent major/minor/angle or independent-tensor fields. `responsivity` is not part of the canonical scientific schema. Historical compatibility metadata may carry the name, but its presence conveys no authority, its absence is not an incomplete APT, and new production does not fabricate a value.

APT-level lineage resolves the observation, calibrator and epoch, source model, fixed nominal beam, TOA amplitude convention, passband/spectrum, source atmosphere, raw-signal definition/sign, WCS/Jacobian, field-rotation law, conventional pivot status, common-origin rule, parent map response, PSF and sensitivity estimator identities, contract/document versions, and immutable parents.

Dense products need not be flattened into each row, but must be immutably resolvable: empirical detector maps, support maps, residuals, full model Jacobians, joint covariance, backgrounds, adequacy diagnostics, broadening and wing constraints, off-source scan membership and statistics, convergence and prior-route records, and complete QC/cause state. An APT without this required lineage is not a complete SCI-BEAM result.

9 Validation and open scientific decisions

Different evidence answers different questions. Algebraic checks can establish the identities, units, coordinate/tensor transformations, atmosphere composition, and state logic in the formal contract. A later code audit and tests can establish implementation conformance. Map-response, WCS, source, atmosphere, and covariance checks are needed for representation fidelity. Synthetic and observational campaigns are needed

Table 1: Examples of independent quantity state.

Situation	Still available	Unavailable or separately limited
Source atmosphere missing	empirical map, core tensor, centroid, amplitude	flxscale ; calibration uses
Off-source noise estimate fails	map fit and valid flxscale	sens ; approximate weights
Precise Gaussian core, shallow field	core tensor/FWHM and health cue	wings, complete PSF, complete beam area
Indefinite broadening difference	effective tensor and residual evidence	literal Gaussian smoothing kernel
Aberrated but well-measured response	health diagnosis; possibly robust amplitude	nominal-science or kernel suitability pending impact study

for bias, scatter, uncertainty calibration, hidden wings, and repeatability. Downstream science-impact studies are needed before kernel use, and production readiness also needs owner-approved numerical policies and operations.

The PSF-recovery study must span nominal and broadened cores, rotated ellipticity, arbitrary broadening orientation, asymmetric/aberrated profiles, halos, sidelobes, shoulders, multiple scales, cropping/masking, varied S/N, realistic map extent, correlated noise, backgrounds, and map-response states. It measures bias/scatter in amplitude, centroid, tensor, FWHM, orientation, broadening, residual diagnostics, and wing detectability.

A hidden-response study asks how much power can remain below noise, outside the map, or absorbed by background, and how this biases amplitude, **flxscale**, and **sens**. A kernel-impact study perturbs width, ellipticity, orientation, centroid, circularization, model family, missing wings, and Gaussian versus empirical response through the relevant downstream science path. It must decide whether a Gaussian kernel is sufficient, a regularized empirical kernel is needed, or no reliable kernel is justified.

Geometry validation includes WCS round trips, rotated ellipses, covariance rotation, circular limits, Jacobian perturbations, broadening recovery, origin gauge, conventional-pivot perturbations, detector-specific angle recovery, pointing/science elevation separation, and exact same-APT transfer. This rationale reports no result from any of these studies.

Table 2: Uncertainty budget and typical correlation scope.

Source	Typical scope	Required treatment
Map noise/covariance	detector/pixel and map-process dependent	joint fit covariance or labeled approximation
Source size/shape	source, epoch, array/passband	joint source-beam propagation
WCS/Jacobian/rotation	observation, detector support, frame	coordinate-centroid-shape covariance
Nominal source amplitude	source, epoch, array	shared flxscale systematic/covariance
Source atmosphere	time/support, often array/observation shared	once-only BEAM propagation
Background/model mismatch	detector/support/model dependent	adequacy and amplitude-bias state, not only Hessian error
Off-source noise statistic	detector/scan plus correlated modes	retained scan values, scatter, and declared estimator
Conventional pivot	observation/elevation, partly common	common plus differential geometry uncertainty

Open scientific decision: SCI-BEAM-OD-001 — Sensitivity estimator and support

The exact scan noise statistic, source-exclusion radius, scan admission, exposure/bandwidth normalization, correlated-noise treatment, and uncertainty statistic remain unresolved.

Open scientific decision: SCI-BEAM-OD-002 — Adequacy, wings, and kernels

Evidence must determine diagnostic thresholds, required PSF accuracy for each use, map depth/radius for wing constraints, and the criteria for Gaussian, empirical, or unavailable kernels.

Open scientific decision: SCI-BEAM-OD-003 — Physical pivot registration

A future authority must establish the physical boresight and field-rotation pivot, acceptable conventional-pivot residuals, uncertainty, epoch, and transfer domain.

Bounded references

1. S. Bryan et al., *Optical Design of the TolTEC Millimeter-wave Camera*, 2018, arXiv:1807.00097. Used only for bounded TolTEC optical/instrument context.
2. J. Golec and the TolTEC Collaboration, *Early high-resolution millimeter-wave maps from TolTEC*, 2024, <https://doi.org/10.1051/epjconf/202429300022>. Used only for bounded commissioning and public Beammap-output context.